

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA0201

COURSE OUTCOME COVERED

***CO2:** Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

*I **B. Deepthi (192424269)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Deepthi

Question 1: Salt-and-Pepper Noise Removal Before Edge Detection

a. Identify the most suitable filtering technique

The **Median Filter (3×3)** is the most suitable filter for removing salt-and-pepper noise. It replaces the center pixel with the median value of its neighboring pixels instead of calculating the average. This effectively removes isolated black and white noise pixels while preserving important image edges and fine details. Because edge detection depends on sharp intensity changes, the median filter is commonly used as a preprocessing step.

b. Explain why the selected filter performs better than an averaging filter

The median filter is more effective because it removes impulse noise without changing the actual edge intensity values. In contrast, an averaging filter computes the mean of neighboring pixels, causing noisy pixels to spread into surrounding areas and producing blurred edges. As a result, the median filter preserves object boundaries, improves image quality, and provides more accurate edge detection.

c. Estimate the probability that a noisy center pixel will be corrected

Given

- Image noise density = **0.05**
- Filter size = **3×3**

Assuming the neighboring pixels are mostly noise-free,

$$P = (1 - 0.05)^8 = (0.95)^8$$

$$P \approx 0.663$$

Therefore, the probability that the noisy center pixel is corrected is approximately **66.3%**.

Assumption: At least five of the nine pixels in the window remain uncorrupted, allowing the median value to represent the original pixel correctly.

d. Estimate the total number of noisy pixels

Image size

$$1024 \times 1024 = 1,048,576 \text{ pixels}$$

Noise density = **5%**

$$1,048,576 \times 0.05 = 52,428.8$$

Approximately **52,429 pixels** are affected by salt-and-pepper noise.

e. Estimate noisy pixels remaining after filtering

If the median filter removes approximately **95%** of the noisy pixels,

Remaining noisy pixels

$$52,429 \times 0.05 = 2,621$$

Therefore, only about **2,621 noisy pixels** remain after preprocessing, giving a much cleaner image for edge detection.

f. If Gaussian noise is present instead

Gaussian noise affects almost every pixel by adding random intensity variations rather than isolated black and white pixels. In this case, a **Gaussian filter** or **Wiener filter** is preferred because these filters smooth the random fluctuations while maintaining acceptable image quality. The preprocessing strategy changes from impulse-noise removal to statistical noise reduction before edge detection.

Question 2: Discretization of a Continuous Image

a. Explain sampling and quantization

To convert an analog image into a digital image, two important processes are performed.

Sampling divides the continuous image into equally spaced pixels along the x and y directions.

Quantization assigns each sampled pixel an intensity value selected from a fixed number of gray levels.

Together, these processes allow the image to be stored, transmitted, and processed by computers.

b. Spatial variation pattern

The image intensity varies according to a **sinusoidal wave** along the x-axis. This means the brightness continuously increases and decreases in a smooth periodic pattern. Along the y-axis, the intensity remains constant.

c. Minimum sampling requirement

According to the **Nyquist Sampling Theorem**, the sampling frequency must be at least twice the highest frequency contained in the image.

$$f_s \geq 2f$$

This prevents information loss during digitization.

d. What happens if sampled below Nyquist rate?

Sampling below the required rate causes **aliasing**, where different image patterns become indistinguishable. This results in jagged edges, false textures, distortion, and loss of fine image details, making further processing inaccurate.

e. Suitable 8-bit quantization

An **8-bit quantization** scheme uses **256 gray levels** ranging from 0 (black) to 255 (white). This provides sufficient intensity resolution for most grayscale image-processing applications while keeping storage requirements reasonable.

f. Effect of quantization

Quantization reduces storage space but introduces a small loss in intensity accuracy because continuous values are rounded to the nearest available gray level. Higher bit depth improves image quality but increases memory usage. An 8-bit image provides a good balance between visual quality and storage efficiency.

Question 3: Edge Detection in Thin Line Images

a. Compare Sobel, Prewitt and Canny

Sobel uses weighted gradient masks and provides better noise resistance than Prewitt.

Prewitt is computationally simpler but is more sensitive to image noise and produces less accurate edges.

Canny combines Gaussian smoothing, gradient calculation, non-maximum suppression, and double thresholding, producing the most accurate and continuous edges. It is the preferred method for detecting very thin lines.

b. Compute Sobel gradient response

Standard Sobel operator

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$
$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

For a horizontal edge,

$$G_x = 0, G_y = 1020$$

Gradient magnitude

$$G = \sqrt{G_x^2 + G_y^2} = 1020$$

The large gradient magnitude indicates a strong edge.

c. Role of gradient magnitude and direction

The **gradient magnitude** measures the strength of the edge. Higher values indicate stronger intensity changes.

The **gradient direction** identifies the orientation of the edge, such as horizontal, vertical, or diagonal. Together, these values help detect object boundaries accurately.

d. Role of Gaussian smoothing

Gaussian smoothing removes random noise before gradient computation. This reduces false edge detection and improves the stability of the Canny algorithm. It ensures that only meaningful intensity changes are considered during edge detection.

e. Effect on edge localization

Although Gaussian smoothing reduces noise, excessive smoothing spreads edge intensity across neighboring pixels. This slightly shifts edge positions and may remove extremely thin lines. Therefore, selecting an appropriate Gaussian kernel size is important.

f. Effect of reducing Canny threshold

Reducing the threshold makes the detector more sensitive.

- More weak edges are detected.
- More image details become visible.
- Noise may also be detected as edges.
- False positives increase.

Thus, threshold selection is a trade-off between edge completeness and noise suppression.

Question 4: Feature Extraction for Handwritten Digit Recognition

a. Working principles

SIFT (Scale-Invariant Feature Transform) detects stable keypoints over different scales and computes highly distinctive descriptors that remain reliable under scaling and rotation.

SURF (Speeded-Up Robust Features) improves processing speed by approximating the Hessian matrix while maintaining good robustness.

ORB (Oriented FAST and Rotated BRIEF) combines FAST keypoint detection with BRIEF descriptors, making it suitable for real-time systems because of its low computational cost.

b. Comparison

Feature	SIFT	SURF	ORB
Scale invariant	Yes	Yes	Partial
Rotation invariant	Yes	Yes	Yes
Processing speed	Slow	Medium	Fast
Computational cost	High	Medium	Low
Real-time suitability	Moderate	Good	Excellent

c. Rotated by 45°

Since SIFT, SURF, and ORB all include rotation handling, they can still detect corresponding feature points after a 45° rotation. SIFT generally provides the most reliable matching, while ORB offers faster performance.

d. Compute total keypoints

Average keypoints = **120**

Images = **50**

$$120 \times 50 = 6000$$

Total keypoints detected = **6,000**.

e. Suitable dimensionality reduction

Principal Component Analysis (PCA) is recommended because it transforms the original features into a smaller set of principal components while preserving approximately **95%** of the useful information. This reduces computation without significantly affecting recognition accuracy.

f. Benefits of dimensionality reduction

Dimensionality reduction decreases memory usage, reduces computational complexity, speeds up classifier training, minimizes redundant information, and lowers the risk of overfitting. As a result, the handwritten digit recognition system becomes faster and more accurate.

Question 5: Morphological Preprocessing for Character Recognition

a. Remove white noise dots

The **Opening** operation (Erosion followed by Dilation) removes isolated white pixels from the background while preserving the main character shapes.

b. Fill small gaps

The **Closing** operation (Dilation followed by Erosion) fills small holes and broken gaps inside character strokes, improving character connectivity.

c. Difference between Opening and Closing

Opening removes small foreground objects and smooths outer boundaries.

Closing fills small holes, connects broken strokes, and smooths inner boundaries.

Both operations improve image quality before feature extraction.

d. Effect of a 3×3 structuring element

A 3×3 square structuring element effectively removes tiny noise while preserving most handwritten strokes. Very thin strokes may become slightly thinner, but overall character readability remains good.

e. Risk of using a large structuring element

A large structuring element may remove important handwriting details, merge neighboring characters, distort stroke shapes, and reduce recognition accuracy. Therefore, the structuring element should be selected carefully based on character size.

f. Complete preprocessing pipeline

1. Acquire the handwritten image using a scanner or camera.
2. Convert the image to grayscale.
3. Apply thresholding to create a binary image.
4. Remove background noise using the Opening operation.
5. Fill broken strokes using the Closing operation.
6. Normalize image size and orientation.
7. Perform thinning or skeletonization if required.
8. Extract features using SIFT, ORB, HOG, or similar methods.
9. Apply dimensionality reduction (PCA).
10. Feed the extracted features into a classifier such as SVM, KNN, CNN, or Neural Network for final character recognition.

Conclusion

Image preprocessing is a fundamental step in digital image processing and computer vision. Techniques such as median filtering, sampling, quantization, edge detection, feature extraction, and morphological operations improve image quality by reducing noise, preserving important details, and enhancing feature representation. These preprocessing methods increase the accuracy, speed, and reliability of computer vision applications such as surveillance systems, medical imaging, document analysis, optical character recognition (OCR), autonomous vehicles, and handwritten digit recognition. Proper selection of preprocessing techniques ensures efficient classification and better overall system performance.